

ECN 712 Fall 2025
EE Schlee Problem Set I
Due Thursday, 28 August

Write in complete, to-the-point sentences.

1. Prove part (iii) of Proposition 1.B.19 on page 7 of Mas-colell, Whinston, and Green (henceforth MWG). Does it continue to hold if completeness is dropped?
2. A real-valued function $u : X \rightarrow \mathbb{R}$ represents a binary relation $\succsim \subset X \times X$, if, for every x and y in X ,

$$x \succsim y \text{ if and only if } u(x) \geq u(y).$$

Suppose that u represents $\succsim$. Evaluate: “ v represents $\succsim$ if and only if there is a strictly-increasing function $f : \text{Range}(u) \rightarrow \mathbb{R}$ such that $v(x) = f(u(x))$ for every $x \in X$.” Prove your assertion.

3. Give a ‘non-pathological’ concrete example of a pair of goods for which conditional preferences over them are *plausibly* not convex. (Define *conditional preferences* over a pair of goods by fixing the consumption levels of the remaining goods.) Assume that the consumption set X is $\mathbb{R}_+^L$.
4. *Homotheticity*. For the consumption set $X = \mathbb{R}_+^L$, let $\succsim \subset X \times X$ be a complete, transitive, continuous, and monotonic binary relation. Prove that it is *homothetic* (MWG, p. 45) if and only if there is a homogeneous of degree 1 representation of it.¹ (Suggestion: For the “only if” part consider the representation u of a monotonic preference relation discussed during our second meeting: $u(x)\mathbf{e} \sim x$ for all $x \in \mathbb{R}_+^L$, where $\mathbf{e}$ is a vector of 1’s.)
5. *Quasilinearity*. This problem is drawn from Kreps, p. 48, Problem 2.11 and MWG, 3.C.5(b). Let the consumption set be $X = Y \times \mathbb{R}$, where $\emptyset \neq Y \subset \mathbb{R}_+^{L-1}$. Write $x \in X$ as $x = (y, m)$ where $y \in Y$ and $m \in \mathbb{R}$. A complete, transitive binary relation $\succsim \subset X \times X$ is *quasilinear with respect to good L* if²
 - (i) For every $y \in Y$ and m and m' in $\mathbb{R}$, $(y, m') \succsim (y, m)$ if and only if $m' \geq m$;
 - (ii) For every y and y' in Y and m and m' in $\mathbb{R}$,
 $(y', m') \succsim (y, m)$ if and only if $(y', m' + \delta) \succsim (y, m + \delta)$ for every $\delta \in \mathbb{R}$; and
 - (iii) For every y and y' in Y there are real numbers m and m' such that $(y, m) \sim (y', m')$.

Prove that $\succsim \subset X \times X$ is quasilinear if it is represented by a function of the form $u(y, m) = \phi(y) + m$ for all $(y, m) \in X$. Illustrate condition (ii) in a diagram and give a verbal interpretation. (If you have the time, I suggest that you think seriously about the “only if” direction, the more delicate part of the argument.)

6. *Composite Commodity Theorem*. Consider MWG, page 99, exercise 3.5.G, part (a). Show that (x^*, z^*) solves the problem described in part (a) if, and only if, there is a solution (x', y') to the full consumer problem with $x' = x^*$ and $\alpha q_0 \cdot y' = \alpha z^*$. (The single commodity ‘ z ’ is a composite of the goods in the vector y . For those of you who know dynamic programming, the idea here should be familiar.)

My part (b): Applications typically aggregate goods into composites, for example “total consumption expenditure in a quarter (of a year).” To what extent does this result justify such a practice? Do you see any qualifications to be made in such a justification?

¹MWG’s definition of homothetic preferences is not standard. The usual definition is for the weak preference relation: “For every x and y in X , $x \succsim y$ if and only if $\lambda x \succsim \lambda y$ for every $\lambda \geq 0$.” I suggest you think about the relationship between the two definitions. Does either imply the other? Can you find additional conditions under which they are equivalent?

²I copy out Kreps’s definition of quasilinear preferences, but I follow MWG’s assumption on the consumption set; in particular MWG allow the good that preferences are quasilinear in to be any real number. I would call (i) strict monotonicity in good L ; and (iii) solvability. (ii) is really the heart of the matter; (iii) is a bit of a shortcut. If continuity is added to this ensemble of assumptions, (iii) follows from (i) and (ii). (i) permits $\succsim$ to satisfy LNS even if the set Y is a finite set, allowing for indivisibilities.